Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The function $f(x) = \sin \frac{\pi x}{n!} - \cos \frac{\pi x}{(n+1)!}$ is

(A) non periodic

(B) periodic with period $2(n!)$

(C) periodic with period $2(n+1)!$

(D) periodic with period $n!$

Q.2

Let $f(\theta) = \begin{vmatrix} \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \end{vmatrix}$ then f is

(A) a non periodic function

(B) periodic with period π

(C) periodic with period $\pi/2$

(D) odd function

Q.3

If $f(x) = \sin x - \cos x$ is written as $f_1(x) + f_2(x)$ where $f_1(x)$ is even and $f_2(x)$ is odd then

(A) $f_1(x) = \cos x$

(B) $f_1(x) = -\cos x$

(C) $f_2(x) = -\sin x + \cos x$

(D) $f_2(x) = \sin(2\pi - x)$

Q.4

If f is a function of real variable x satisfying $f(x+4) - f(x+2) + f(x) = 0$, then f is periodic with period

(A) 8

(B) 10

(C) 12

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(D) 6

Q.5

Let $f(x) = \sin \sqrt{x}$, then

(A) $f(x)$ is periodic with period $\sqrt{2\pi}$

(B) $f(x)$ is periodic with period $\sqrt{\pi}$

(C) $f(x)$ is periodic with period $4\pi^2$

(D) none of these

Q.6

Which of the following functions is nonperiodic

(A) $f(x) = x - [x]$

(B) $f(x) = \begin{cases} 1 \text{ if } x \text{ is a rational number} \\ 0 \text{ if } x \text{ is an irrational number} \end{cases}$

(C) $f(x) = \sqrt{\frac{8}{1+\cos x} + \frac{8}{1-\cos x}}$

(D) none of these

Q.7

The value of $n \in \mathbf{z}$ for which the function $f(x) = \frac{\sin nx}{\sin(x/n)}$ has 4π as its period is

(A) 2

(B) 3

(C) 4

(D) 5

Q.8

Let $f: R \to R$ be a periodic function such that $fT + x = 1 + 1 - 3fx + 3fx^2 - fx^{3^{\frac{1}{3}}}$ where T is a fixed positive number, then period of fx is

(A) T

(B) $2T$

(C) $3T$

(D) None of these

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Q.9

For the function $fx = \sin\pi x \times \cos^{-1}x$, choose the correct option.

(where . represents the greatest integer function)

(A) Domain of $f(x) \in [-1,1]$

(B) Range of $f(x)$ contains exactly two elements

(C) $f(x)$ is an identity function

(D) $f(x)$ is a constant function

Q.10

Period of the function $\left|\sin^3 \frac{x}{2}\right| + \left|\cos^5 \frac{x}{5}\right|$ is

(A) 2π

(B) 10π

(C) 8π

(D) 5π

Q.11

If $\mathrm{f} : [-20, 20] \to \mathrm{R}$, defined by $\mathrm{f(x)} = \left[\frac{x^2}{a}\right] \sin \mathrm{x} + \cos \mathrm{x}$, (where [.] denotes the greatest interger function) is an even function, then the set of values of $'\mathrm{a}^n$ is given by

(A) null set

(B) R

(C) $[0, 400]$

(D) $(400, \infty)$

Q.12

Which of the following functions is even,

(A) $f(x) = \sqrt{1 + x + x^2} - \sqrt{1 - x + x^2}$

(B) $f(x) = \log\left(\frac{1-x}{1+x}\right)$

(C) $f(x) = \log\left(x + \sqrt{1 + x^2}\right)$

(D) $f(x) = \frac{e^x + e^{-x}}{2}$

Q.13

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$f(x) = \left(\sin x^7\right) \cdot e^{x^5 \operatorname{sgn} x^9}$ is

(A) an even function

(B) an odd function

(C) neither even nor odd

(D) None of these

Q.14

Consider the following statements. Let $\mathrm{A} = \{1, 2, 3, 4\}$ and $\mathrm{B} = \{5, 7, 9\}$

$\mathrm{A} \times \mathrm{B} = \mathrm{B} \times \mathrm{A}$

II. $\mathrm{n}(\mathrm{A} \times \mathrm{B}) = \mathrm{n}(\mathrm{B} \times \mathrm{A})$

Choose the correct option.

(A) Statement-I is true.

(B) Statement-II is true.

(C) Both are true.

(D) Both are false.

Q.15

Which pairs of function is identical?

(A) $f(x) = \sqrt{x^2}, g(x) = x$

(B) $f(x) = \sin^2 x + \cos^2 x; g(x) = 1$

(C) $f(x) = \frac{x}{x}, g(x) = 1$

(D) None of these

Answers & Solutions

Q.1 Answer:

periodic with period $2(n+1)!$

Solution:

Since the period of $\sin x$ **is** 2π **so the period of** $\sin \frac{\pi x}{n!}$ **is** $\frac{2\pi n!}{\pi} = 2(n!)$.
The period of $\cos \frac{\pi x}{(n+1)!}$ **is**
$2(n+1)!$ **The period of** $f(x) =$ **l.c.m.** $(2(n!), 2(n+1)!) = 2(n+1)!$

Q.2 Answer:

periodic with period $\pi/2$

Solution:

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$$f(\theta)=\begin{vmatrix} -1 & 0 & 1 \\ -1 & 1 & 0 \\ 1+\sin^2\theta & \cos^2\theta & 4\sin 2\theta \end{vmatrix}$$

$(R_1 \to R_1 - R_3$

$R_2 \to R_2 - R_3)$

$$=\begin{vmatrix} 0 & 0 & 1 \\ -1 & 1 & 0 \\ 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \end{vmatrix} \quad (C_1 \to C_1 + C_3)$$

$$= -\left(\cos^2\theta + 1 + \sin^2\theta + 4\sin 4\theta\right)$$

$$= -2(1 + 2\sin 4\theta)$$

which is periodic function with period $\frac{2\pi}{4} = \frac{\pi}{2}$.

Q.3 Answer:

$f_1(x) = -\cos x$

Solution:

$f_1(x) = \frac{f(x)+f(-x)}{2} = \frac{1}{2}[\sin x - \cos x - \sin x - \cos x]$

$= -\cos x$

$f_2(x) = \frac{f(x)-f(-x)}{2} = \frac{1}{2}[\sin x - \cos x + \sin x + \cos x]$

$= \sin x$

Q.4 Answer:

12

Solution:

$f(x+4) - f(x+2) + f(x) = 0 \forall x \in \mathbf{R}$

Replace x by $x+2$,

$f(x+6) - f(x+4) + f(x+2) = 0$

Adding above two equations, we have

$f(x+6) + f(x) = 0 \forall x \in \mathbf{R}$

Replace x by $x+6$, we get

$f(x+12) + f(x+6) = 0$

$\Rightarrow f(x+12) = -f(x+6) = f(x)$

f is periodic with period 12.

Q.5 Answer:

none of these

Solution:

Since $f(x+T) = \sin\sqrt{x+T} \neq \sin\sqrt{x}$ for any value T.

$f(x)$ is not periodic.

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Q.6 Answer:

none of these

Solution:

The function in (a) is periodic with period 1 and the function in (b) is also periodic since $f(x+r)=f(x)$ for every rational r. The function in (c) is equal to $\frac{4}{|\sin x|}$ and thus has period π.

Q.7 Answer:

2

Solution:

$f(x+4\pi)=f(x)$

If $\frac{\sin n(x+4\pi)}{\sin \frac{x+4\pi}{n}}=\frac{\sin nx}{\sin \frac{x}{n}}$

which holds if $n=2.$

Q.8 Answer:

$2T$

Solution:

Given : $fT+x=1+1-fx^{3\frac{1}{3}}$

$=1+1-fx$

$\Rightarrow fT+x+fx=2$ **... (1)**

$\Rightarrow f2T+x+fT+x=2$ **... (2)**

Subtracting (1) from (2)

$f2T+x-fx=0$

$f2T+x-fx=0$

Also, T **is positive therefore period of** $fx=2T$

Q.9 Answer:

fx is a constant function

Solution:

For any $x \in R$**, we know,**

$\sin\pi x=0$

$\therefore fx=0, \quad \forall x \in -1,2 \because -1 \leq x \leq 1 \quad \text{for} \quad \cos^{-1}x \quad \text{to} \quad \text{be} \quad \text{defined}$

$\therefore$ **Domain** $x \in -1,2$

$\Rightarrow$ **Range** $=0$

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$\therefore fx$ **is a constant function**

Q.10 Answer:

10π

Solution:

Period of $\sin x = 2\pi$
$\Rightarrow$ **period of** $\sin^3 x = 2\pi$
period of $\left|\sin^3 x\right| = \pi \Rightarrow$ **period of** $\left|\sin^3 \frac{x}{2}\right| = 2\pi$
period of $\cos^5 x = 2\pi \Rightarrow$ **period of** $\left|\cos^5 x\right| = \pi$
$\Rightarrow$ **period of** $\left|\cos^5 \frac{x}{5}\right| = 5\pi$
Thus required period = LCM of $2\pi \& 5\pi = 10\pi$

Q.11 Answer:

$(400, \infty)$

Solution:

Clearly, $\mathrm{f(x)}$ **will be an even number,**
if $\left[\frac{x^2}{a}\right] = 0$
$\forall \mathrm{x} \in [-20, 20]$
$\therefore 0 \leq \frac{x^2}{a} < 1$
$\therefore 0 \leq x^2 < 400$
$\Rightarrow \quad a > 400$

Q.12 Answer:

$f(x) = \frac{e^x + e^{-x}}{2}$

Solution:

$f(-x) = \sqrt{1 + (-x) + (-x)^2} - \sqrt{1 - (-x) + (-x)^2}$
$= \sqrt{1 - x + x^2} - \sqrt{1 + x + x^2} = -f(x)$
Hence, $f(x)$ **is odd.**
(b) $f(-x) = \log\left\{\frac{1-(-x)}{1+(-x)}\right\} = \log\left(\frac{1+x}{1-x}\right) = -f(x)$
Hence, $f(x)$ **is odd.**
(c) $f(-x) = \log\left(-x + \sqrt{1 + (-x)^2}\right)$
$= \log\left\{\frac{\left(-x+\sqrt{1+x^2}\right)\left(x+\sqrt{1+x^2}\right)}{\left(x+\sqrt{1+x^2}\right)}\right\}$
$= \log\left(\frac{1}{x+\sqrt{1+x^2}}\right) = -f(x)$
Hence, $f(x)$ **is odd.**

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(d) $f(-x)=\frac{e^{-x}+e^{-(-x)}}{2}=\frac{e^{-x}+e^{x}}{2}=f(x)$

Hence, $f(x)$ **is even.**

Q.13 Answer:

an odd function

Solution:

$f(x)=\left(\sin x^{7}\right) e_{0}^{x^{5} \operatorname{sgn} x^{9}}$

$\sin x^{7} \rightarrow$ **odd,** $\operatorname{sgn} x^{9} \rightarrow$ **odd,** $x^{5} \rightarrow$ **odd**

$\therefore \quad e^{x^{5} \operatorname{sgn} x^{9}} \rightarrow$ **even**

$\therefore \quad f(x)=\text{odd} \times$ **even** $=$ **odd**

Q.14 Answer:

Statement-II is true.

Solution:

$A \times B=\{(1,5),(1,7),(1,9),(2,5),(2,7),(2,9)$
$(3,5),(3,7),(3,9),(4,5),(4,7),(4,9)\}$

$$B \times A=\left\{\begin{array}{l}(5,1),(5,2),(5,3),(5,4) \\ (7,1),(7,2),(7,3),(7,4) \\ (9,1)(9,2),(9,3),(9,4)\end{array}\right\}$$

$A \times B \neq B \times A$ **but** $n(A \times B)=n(B \times A)=12$

Q.15 Answer:

$f(x)=\sin^{2} x+\cos^{2} x ; g(x)=1$

Solution:

For checking equal function

(a) Domain of $f(x)=\mathrm{R}$ **but range** $=[0, \infty)$ **Domain of** $g(x)=\mathrm{R}$**, range** $=\mathrm{R}$ **Domain same but range is different so it is not an equal function.**

(b) Domain of $f(x)=\mathrm{R}$ **Domain of** $g(x)=\mathrm{R}$ **Domain and range both same so it is an equal function.**

(c) Domain of $f(x)=\mathrm{R}-\{0\}$ **Domain of** $g(x)=\mathrm{R}$

Not equal function as domain is different.

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